

Practical 1

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Electricity and Electronics EBN 122

Semester 2, 2023

Practical Information

Maximum marks:	200	Full marks:	200
Duration of practical:	180 min	Open/Closed book:	Open
Total number of pages (this page included)		14	

Lecturers: Prof P. de Villiers, Prof P.J. Jacobs**Assistant Lecturers:** N.-J. Lord, G. Neeuwfan, M. Modiba**Overview of Practicals**

The engineering process to build any circuit consists of three parts:

- **Calculations**

A circuit design is proposed. The circuit is analyzed using fundamental circuit theory, e.g., Ohm's Law, Kirchoff's Laws, Nodal analysis, Mesh analysis etc. Calculations are done by hand to determine specific voltages and/or currents in the circuit. This is the first step to ensure the proposed circuit design is correct.

- **Simulation**

The proposed circuit is simulated using software such as LTspice. The simulation includes non-ideal effects which are not included in the calculations. The simulation results should coincide closely with the calculations done previously. The simulation allows us to analyze the circuit for a wide variety of circumstances. The simulations verify that the proposed circuit will perform its intended function.

- **Laboratory/Practical**

The proposed circuit is built using physical components. The hardware is connected to an actual power source such as a battery. Laboratory equipment can be used to measure the voltages or currents determined in the calculation and simulation steps above. The measurements should coincide closely with the results in the *Calculation* step. It will not be exact as there are tolerances in the physical components which are not included in the *Calculations*. The measurements should also be similar to the simulations, but will not be exactly the same since not all non-ideal characteristics are included in the simulation.

Practicals Process

The EBN 122 Practical will follow the engineering process discussed above. As shown in the figure below, a complete Practical consists of a **Pre-Practical**, a **Demonstration**, and a **Laboratory** part. The Laboratory part is also often referred to as a 'Practical session'.

1. Section 1: Pre-Practical - Calculations and Simulation [60 marks]

- Analyse the circuit diagrams in the practical guide using the theory covered in EBN 122. *Calculate* the various unknowns.
- *Simulate* the circuits using LTSpice and compare the results to your calculations.
- Download the Excel spreadsheet for your Pre-Practical on AMS and fill in your answers. Submit your Excel spreadsheet on AMS before the relevant deadline.
- You will have to work through the guideline documentation to familiarise yourself with the circuit components and software that are required for the pre-practical.

Pre-Practicals are completed at home before a Practical session.

2. Section 2: Practical Demonstration [20 marks]

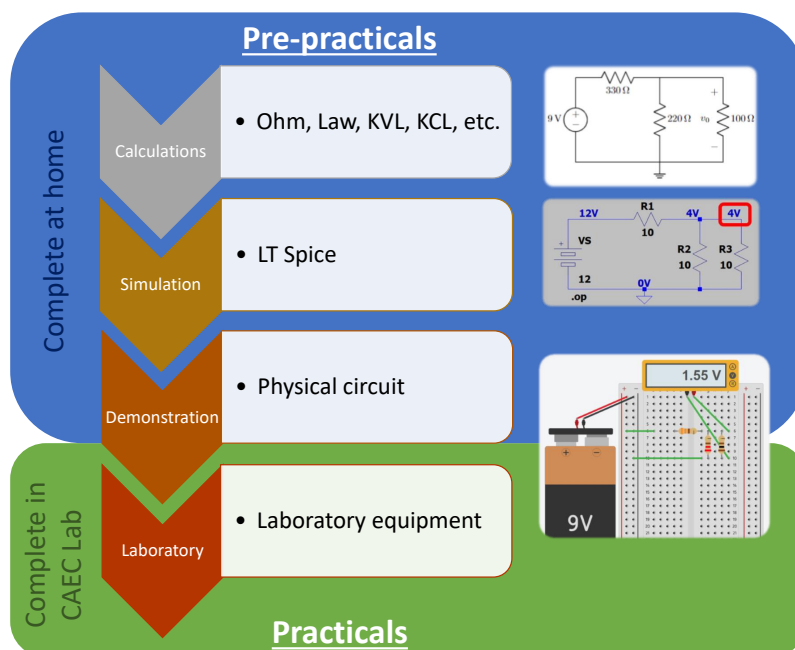
- You will be given a circuit similar to a circuit you analyzed and simulated to build on a breadboard using the component kit provided for EBN 122. You will demonstrate this physical circuit upon entering the laboratory. We will award a mark for the circuit.

3. Section 3: Practical Measurements [120 marks]

- Use laboratory equipment to measure voltages and currents as specified in the practical guide for the circuit you built. Compare it to your Pre-Practical results.
- Download the Excel spreadsheet for your Practical on AMS and fill in your answers. Submit your Excel spreadsheet on AMS before the end of the practical session.

Section 3 is completed in the CAEC Lab during a 3-hour Practical session.

All measurements must be made and submitted during the Practical session.



Assessment evaluation, submission rules and time-line

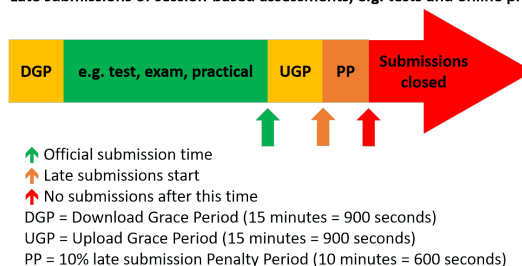
The following guidelines apply when filling in answers on an Excel spreadsheet.

- The Excel spreadsheet for a Pre-Practical or Practical are found on the AMS system. The spreadsheet contains unique parameter values for each student. Therefore, you must download the Excel spreadsheet from your own AMS account. Follow the instructions on the AMS spreadsheet carefully when filling in your answers.
- Final answers must be written as real numbers and not as fractions. Use at least 5 significant figures to write irrational numbers. Use decimal points and NOT decimal commas.
- Answers must include units where applicable. (Refer to the EECE Rules for Electronic Assessments of tests and examinations for guidelines on how to enter units. Also refer to the example list of units on the Excel spreadsheet.)

After filling in your answers an Excel spreadsheet for a Pre-Practical or Practical, you need to upload the Excel spreadsheet to AMS for that assessment to be marked. Each Pre-practical and Practical has a start time and an end-time as indicated on AMS. The following rules and time-line apply for all assessments for EBN 122 as illustrated by the figure below:

1. The Download Grace Period (DGP) is a 15 minute period before the official start of the assessment for students to download all documents related to the assessment and read all the necessary instructions.
2. The start of the green section after the DGP indicates the official start of the assessment.
3. The upward green arrow at the end of the green section indicates the official submission deadline for the assessment.
4. Students must aim to submit their Excel spreadsheet with their answers on the AMS before the official submission deadline.
5. Students can request assistance via **e-mail** ebn2023.practicals@gmail.com if they experience any technical issues with respect to uploading their Excel spreadsheets on the AMS.
6. An Upload Grace Period (UGP) of 15 minutes follows the official submission deadline. Students may submit their Excel spreadsheets with answers during this period without incurring any penalties. No email submissions will be accepted during the UGP.
7. A Penalty Period (PP) of 10 minutes follows the UGP. If a student submits an Excel spreadsheets during this period, the student incurs a 10% penalty to their final assessment mark.
8. No late submissions are accepted after the PP.

Late submissions of session-based assessments, e.g. tests and online practicals



By uploading an Excel spreadsheet to AMS, you declare the following:

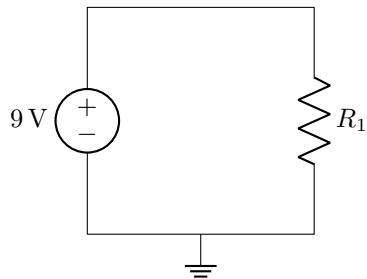
The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University, more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, examinations and/or any other forms of assessment.

Section 1: Pre-Practical - Calculations and Simulation

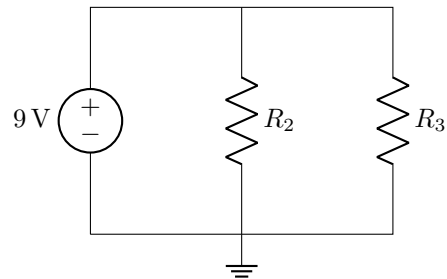
Questions 1 to 10 [10]

Given: R_1 , R_2 , R_3

Circuit 1



Circuit 2



Complete the questions below through theoretical analysis of **Circuit 1** and **Circuit 2**.

01 Calculate the equivalent resistance R_{eq} of **Circuit 2** as seen by the source. [1]

02 Calculate the power P_{R1} dissipated by R_1 . [1]

03 Calculate the power P_{R2} dissipated by R_2 . [1]

04 Calculate the power P_{R3} dissipated by R_3 . [1]

05 Calculate the total power dissipated by the resistors in **Circuit 2**. [1]

Build **Circuit 1** and **Circuit 2** in LTspice and perform a **DC operating point** simulation. Complete the questions below.

06 Measure the power P_{R1} dissipated by R_1 in simulation. [1]

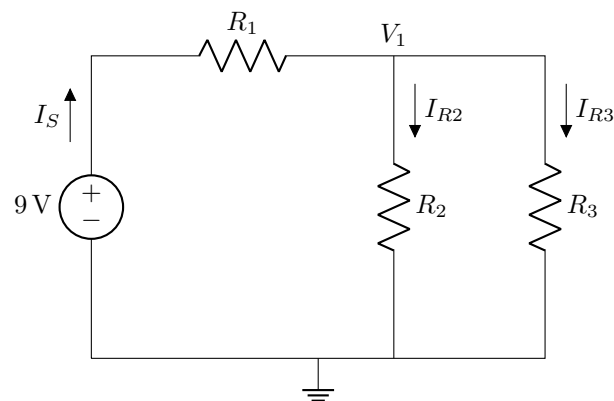
07 Measure the power P_{R2} dissipated by R_2 in simulation. [1]

08 Measure the power P_{R3} dissipated by R_3 in simulation. [1]

09 What is the total power dissipated by the resistors in **Circuit 2** in simulation? [1]

10 What is the maximum power that a single physical resistor in the component kits provided to you can dissipate? [1]

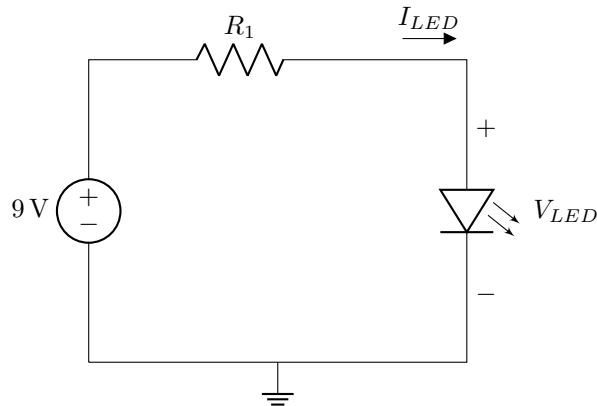
Questions 11 to 20 [10]

Given: R_1 , R_2 , R_3 **Circuit 3**Complete the questions below through theoretical analysis of **Circuit 3**.

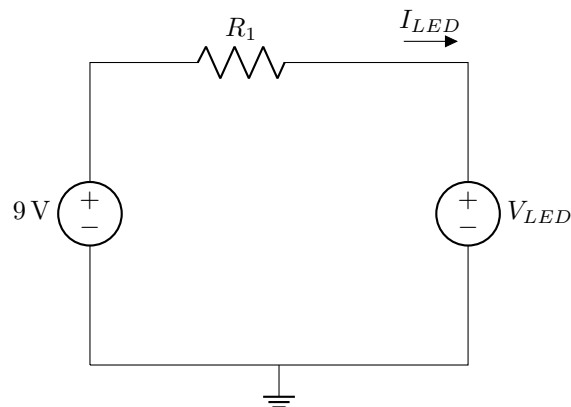
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| 11 | Calculate the voltage at node V_1 . [1] |
| 12 | Calculate I_{R3} . [1] |
| 13 | Calculate the power P_{R3} dissipated by R_3 . [1] |
| 14 | Calculate I_S . [1] |
| 15 | Calculate the equivalent resistance R_{eq} of the circuit as seen by the source. [1] |

Build **Circuit 3** in LTspice and perform a **DC operating point** simulation. Complete the questions below.

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|-----------|---|
| 16 | Measure the voltage at node V_1 in simulation. [1] |
| 17 | Measure I_{R3} in simulation. [1] |
| 18 | Measure the power P_{R3} dissipated by R_3 in simulation. [1] |
| 19 | Measure I_S in simulation. [1] |
| 20 | Measure the equivalent resistance R_{eq} of the circuit as seen by the source. Hint: Use the supply voltage and the measured current I_s to calculate the resistance. [1] |

Questions 21 to 25 [7]Given: V_{LED} , I_{LEDmax} **Circuit 4**

The Light Emitting Diode (LED) in **Circuit 4** can be modelled/represented as a DC voltage source. In other words, as shown in **Circuit 5**, the DC voltage source V_{LED} is an equivalent circuit replacement for the LED. The replacement of the LED with its equivalent circuit allows us to do theoretical calculations and simulations.

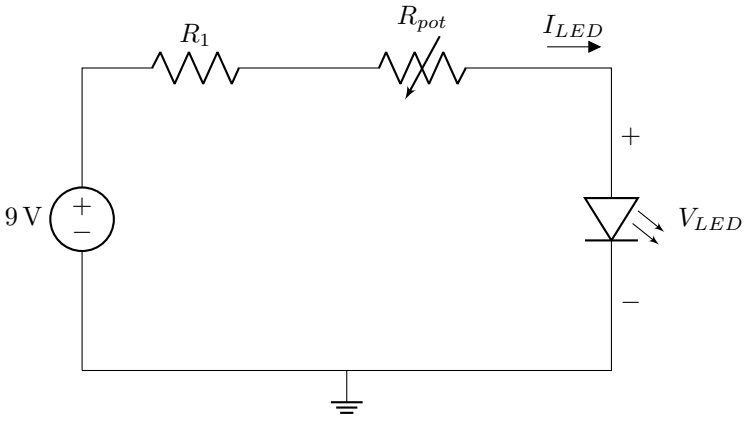
Circuit 5

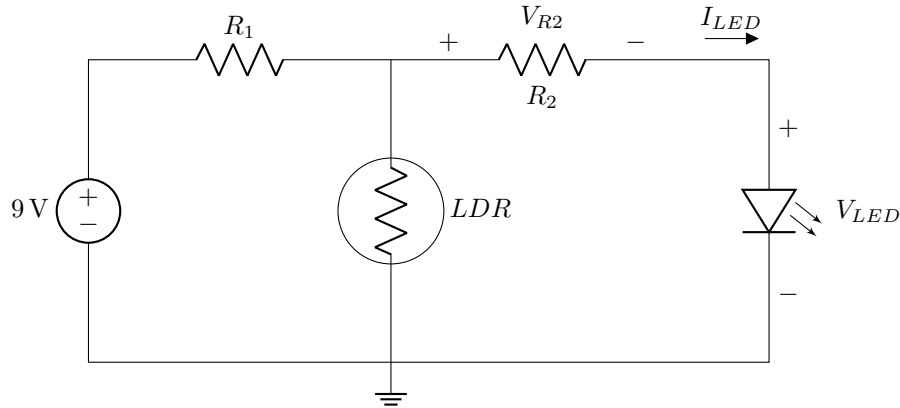
Complete the questions below through theoretical analysis of **Circuit 5**.

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|-----------|--|
| 21 | Calculate the value of R_1 for a maximum current flow of $I_{LED} = I_{LEDmax}$ through the LED. Use this value of R_1 for the four questions below. [2] |
| 22 | Calculate the power P_{LED} dissipated by the voltage source V_{LED} . [1] |

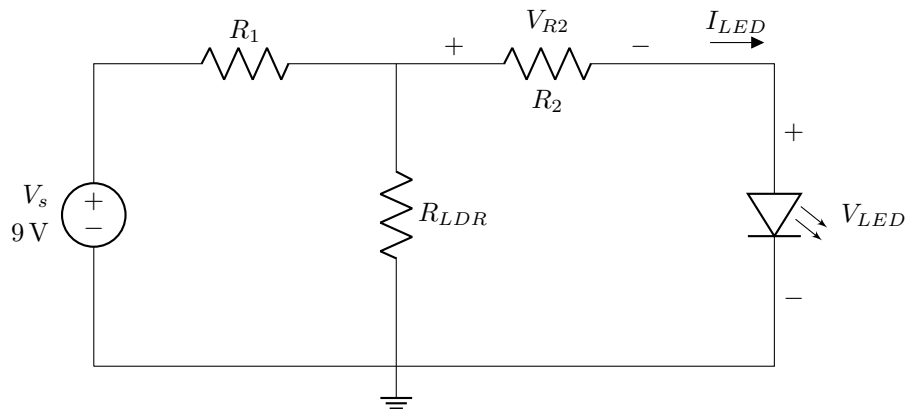
Build **Circuit 5** in LTspice. Perform a **DC operating point** simulation and complete the questions below.

- | | |
|-----------|--|
| 23 | Measure I_{LED} in simulation. [1] |
| 24 | Measure the power P_{LED} dissipated by the LED voltage source V_{LED} in simulation. [1] |
| 25 | The power P_{LED} is an indication of the brightness of the LED. In order to increase the LED brightness, should the value of R_1 increase or decrease? Write 1 for <i>Yes</i> , and write 2 for <i>No</i> . [2] |

Questions 26 to 30 [10]	
Given: R_1 , R_{pot1} , R_{pot2} , R_{pot3}	
Add a potentiometer in your simulation in series with R_1 in Circuit 4 as shown in Circuit 6 . The potentiometer is equivalent to a normal resistor with a variable value. Set the resistance R_{pot} of the potentiometer to the values provided for each question below.	
Circuit 6 	
Build Circuit 6 in LTspice. Use an LED component and not a voltage source. Set the resistance R_{pot} of the potentiometer to the values provided below. Perform a DC operating point simulation and complete the questions below.	
26	Measure the power P_{LED} dissipated by the LED in simulation when $R_{pot} = R_{pot1}$. [2]
27	Measure the power P_{LED} dissipated by the LED in simulation when $R_{pot} = R_{pot2}$. [2]
28	Measure the power P_{LED} dissipated by the LED in simulation when $R_{pot} = R_{pot3}$. [2]
29	What is the maximum power a physical LED as provided in your component kit can dissipate? [2]
Build Circuit 6 on a breadboard using the components provided in the component kit and a 5 k Ω potentiometer. Adjust the potentiometer by turning its shaft/wiper from one extreme to the next. Observe how the LED behaves. Answer the question below.	
30	Does the brightness of the LED increase when the resistance of the potentiometer increases? Write 1 for <i>Yes</i> , and write 2 for <i>No</i> . [2]

Questions 31 to 35 [5]Given: R_1 , R_2 , R_{LDR} **Circuit 7**

The Light Dependent Resistor (LDR) in **Circuit 7** can be modelled/represented as a resistor R_{LDR} as shown in **Circuit 8**.

Circuit 8

Complete the questions below through theoretical analysis of **Circuit 8** with $V_{LED} = 2\text{ V}$.

31 Calculate I_{LED} . [1]

32 Calculate the power P_{LED} dissipated by the LED. [1]

Build **Circuit 8** in LTspice (use an LED, not a voltage source). Perform a **DC operating point** simulation and complete the questions below.

33 Measure V_{LED} in simulation. [1]

34 Measure I_{LED} in simulation. [1]

35 Measure the power P_{LED} dissipated by the LED in simulation. [1]

Questions 36 to 44 [18]

The resistance of the LDR in **Circuit 7** is dependent on the ambient light intensity. The circuit behavior at different light levels can be simulated by sweeping the resistance of resistor R_{LDR} over a range that is typical for a LDR.

Build **Circuit 8** in LTspice. Use the same values for R_1 and R_2 as in Q31 to Q35 above.

Perform a **DC sweep** simulation and sweep R_{LDR} from 10Ω to $5\text{k}\Omega$ while measuring the current I_{LED} and voltage V_{LED} . Complete the questions below using the graph obtained from LTspice.

36	Measure the voltage V_{LEDmax} when the current I_{LED} is maximum. [2]
37	Measure the maximum current I_{LEDmax} through the LED. [2]
38	Measure the maximum power P_{LEDmax} dissipated by the LED. [2]
39	Measure the voltage V_{LEDmin} when the current I_{LED} is minimum. [2]
40	Measure the minimum current I_{LEDmin} through the LED. [2]
41	Measure the minimum power P_{LEDmin} dissipated by the LED in simulation. [2]
42	Measure the voltage V_{LED} when $R_{LDR} = 2.5\text{k}\Omega$. [2]
43	Measure I_{LED} when $R_{LDR} = 2.5\text{k}\Omega$. [2]
44	Measure the power P_{LED} dissipated by the LED when $R_{LDR} = 2.5\text{k}\Omega$ in simulation. [2]

Total Section 1: [60]

Section 2: Practical Demonstration

Questions 45 to 46 [20]

Given: R_1 , R_2

The values for R_1 and R_2 should be same as used for Q31 to Q44 above.

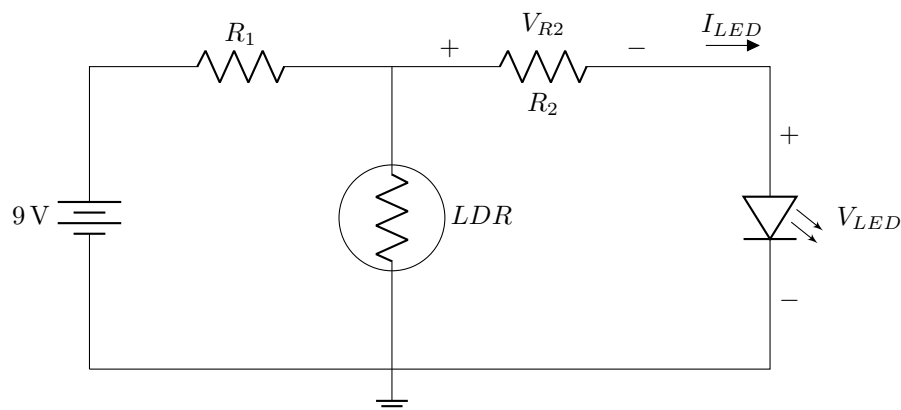
Build **Circuit 9** shown below on a breadboard using the components provided in the component kit. A 9 V battery can be used to power the circuit. This circuit will be used for the practical session. It is advised to read through the next session and build the circuit in such a way that the changes required for the Practical will be easy to implement.

After building the circuit you can test it by placing the circuit in a dark or brightly lit room. The brightness of the LED should change depending on the brightness of the light falling on the LDR.

When entering the lab for your allocated practical session, the circuit will be evaluated by an Assistant Lecturer.

**Please ensure that your circuit is prepared before the practical session.
You will not be allowed to build the circuit during the practical session.**

Circuit 9



- | | |
|-----------|---|
| 45 | Is the LED brightness proportional (write 1) or inversely proportional (write 2) to the brightness of the light falling on the LDR? [2] |
| 46 | Circuit evaluation. [18] |

Total Section 2: [20]

Section 3: Practical Measurements
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Question 47 [5]

Set up the laboratory power supply to supply 9 V DC. Use the multimeter to measure the voltage V_S supplied by the power supply and verify that it is between 8.9 V and 9.1 V.

**The measured supply voltage V_S will be used in the marking of subsequent questions.
Take care to ensure that this step is done correctly.**

Once the supply voltage is set, it should not be changed under any circumstances!

47	Enter the measured supply voltage V_S . [5]
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Questions 48 to 51 [20]

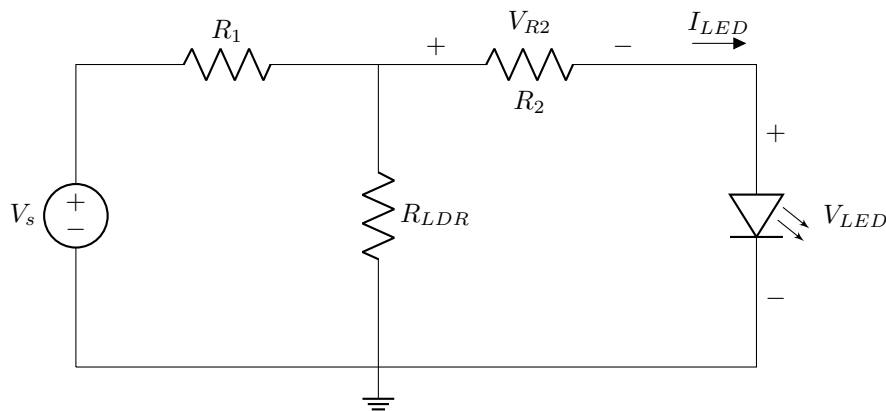
Given: R_1 , R_2 , R_{LDR} .

Important: The values for R_1 , R_2 , and R_{LDR} are the same as for Q31 and Q44 above and will be used throughout the Practical.

Replace the LDR in **Circuit 9** (the circuit that was built for the demonstration) with a fixed resistor of value R_{LDR} . Remove the battery from the circuit and connect the laboratory power supply in its place. The power supply was set to 9 V DC in a previous question and the actual output voltage was measured as V_s .

The circuit should resemble **Circuit 10** shown below.

Use the multimeter to measure the voltages/currents specified below. Remember that current must be measured in series. However, current can be calculated by measuring the voltage across a component first and then using Ohm's law.

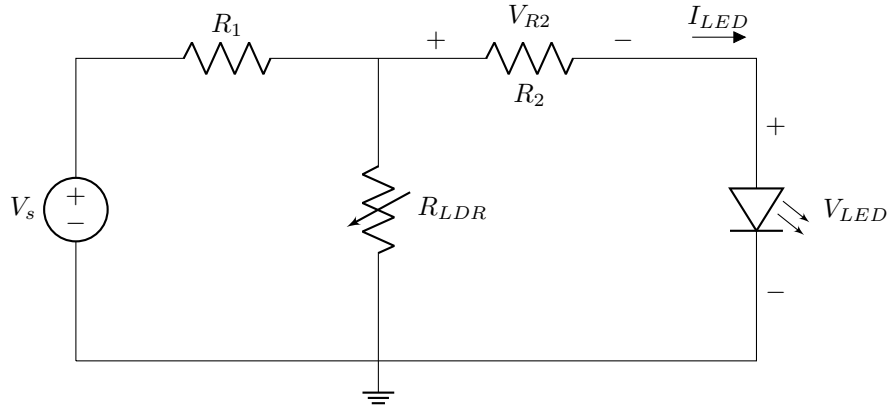
Circuit 10

48	Measure V_{LED} . [5]
49	Measure V_{R2} . [5]
50	Measure I_{LED} . Hint: Use the voltage V_{R2} across R_2 and Ohm's law to calculate I_{LED} . [5]
51	Measure the power P_{LED} dissipated by the LED. Hint: Use V_{LED} and I_{LED} . [5]

Questions 52 to 64 [65]

Replace R_{LDR} in **Circuit 10** with a $5\text{ k}\Omega$ potentiometer as shown in **Circuit 11** below.

Use the multimeter to measure the voltages/currents specified below. Remember that current must be measured in series. However, current can be calculated by measuring the voltage across a component first and then using Ohm's law.

Circuit 11

Turn the potentiometer to where the LED is brightest.

52 Measure V_{LEDmax} . [5]

53 Measure V_{R2} . [5]

54 Measure I_{LEDmax} . Hint: Use V_{R2} and Ohm's law to calculate I_{LED} . [5]

55 Measure the power P_{LEDmax} dissipated by the LED. [5]

Turn the potentiometer where the LED is dimmest.

56 Measure V_{LEDmin} [5]

57 Measure V_{R2} . [5]

58 Measure I_{LEDmin} [5]

59 Measure the power P_{LEDmin} dissipated by the LED. [5]

Turn the shaft/wiper of the potentiometer to approximately the middle of its range. Try to do this as accurately as possible by first finding both extremes of the potentiometer and then moving the shaft/wiper to a position between the extremes.

60 Measure V_{LED} [5]

61 Measure V_{R2} . [5]

62 Measure I_{LED} [5]

63 Measure the power P_{LED} dissipated by the LED. [5]

Take the potentiometer out of the circuit and measure its resistance R_{pot} when the wiper/shaft is in the middle of the range.

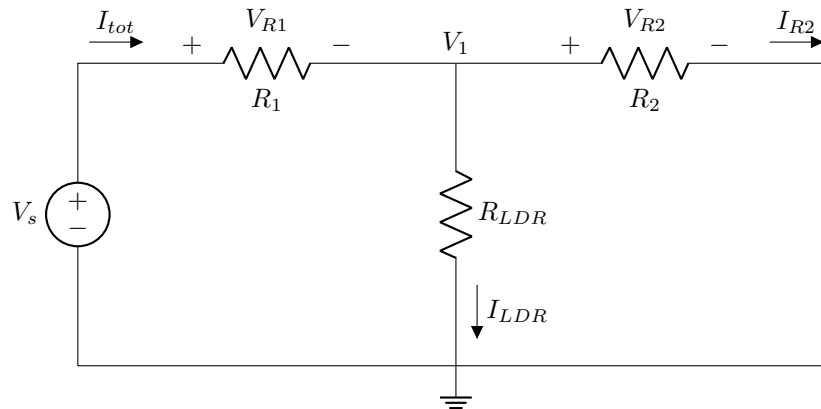
Be careful to not change the wiper/shaft position. Check how it was connected in the circuit and make sure the resistance is measured between the correct legs. Since the human body is also a resistor, take care not to touch the probes or component legs while making resistance measurements.

64 Measure R_{pot} [5]

Questions 65 to 69 [30]

Replace the LED in **Circuit 11** with a wire and the potentiometer with a fixed resistor R_{LDR} . The circuit should look like **Circuit 12** shown below.

Use the multimeter to measure the voltages/currents specified below. Remember that current must be measured in series. However, current can be calculated by measuring the voltage across a component first and then using Ohm's law.

Circuit 12

65	Measure the voltage at node V_1 . [6]
66	Measure I_{R2} . Hint: Use V_1 and Ohm's law. [6]
67	Measure I_{LDR} . Hint: Use V_1 and Ohm's law. [6]
68	Calculate I_{tot} , the total current, using I_{R2} and I_{LDR} . [6]
69	Measure I_{tot} , the total current, using Ohm's law and V_{R1} . [6]

Total Section 3: [120]

Grand Total : [200]